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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

**Title: Analyzing Sales Performance Across Products, Categories,
and Regions Using Power BI**

GitHub Link - [Github Profile Link](#)

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Analyzing Sales Performance Across Products, Categories, and Regions Using Power BI

Problem Statement

Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

Note: The Sample Superstore dataset was specifically used for this analysis, sourced from Kaggle (<https://www.kaggle.com/datasets/naveenkumar20bps1137/sample-superstore>).

Introduction

Retail businesses generate large volumes of transactional data covering products, customers, and regions, but this data is only useful when it can be quickly interpreted by decision-makers. Power BI allows raw transactional records to be transformed into interactive visualizations and key performance indicators, making it possible for management to identify patterns in sales and profitability at a glance. This experiment uses the Sample Superstore dataset to build a set of DAX-driven measures and an interactive multi-page dashboard that highlights business performance across products, categories, and regions.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the Sample Superstore dataset from Kaggle, as specified in the assignment instructions.
- The dataset contains order-level retail transaction records, including product, customer, geographic, and financial details.
- Key fields include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Segment, Country, City, State, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, and Profit.

Step 2: Data Import into Power BI

- Opened Power BI Desktop and used the Get Data option to import the Sample Superstore dataset.
- Loaded the dataset as a single table named "Sample_ Superstore".
- Confirmed all columns loaded correctly in the Data pane, including the base fields and later-added DAX measures.

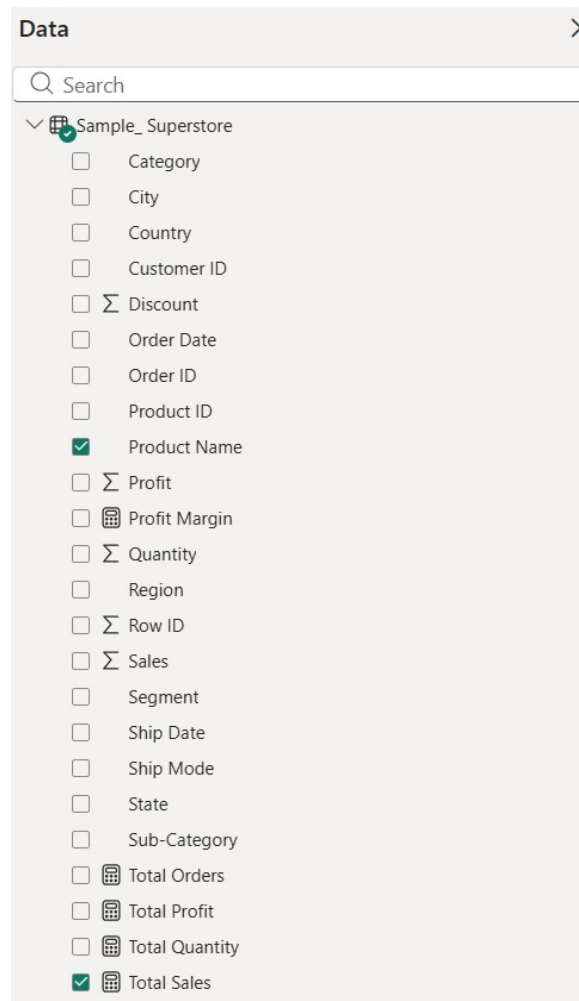


Fig 1: Sample_Superstore fields loaded in the Power BI Data pane

Step 3: Data Exploration and Verification

- Reviewed the loaded table in Data view to verify the data, confirming 9,994 rows were loaded successfully.
- Verified column data types — numeric fields for Sales, Quantity, Discount, and Profit, date fields for Order Date and Ship Date, and text fields for the remaining categorical columns.
- As the dataset is a single flat table, no relationships or data modeling between multiple tables were required.

Table: Sample_Superstore (9,994 rows) Column: Profit Margin (distinct values)

Fig 2: Sample_Superstore table in Data view (9,994 rows)

Step 4: DAX Measures

Created the following DAX measures on the Sample_Superstore table to calculate key business KPIs:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Profit Margin

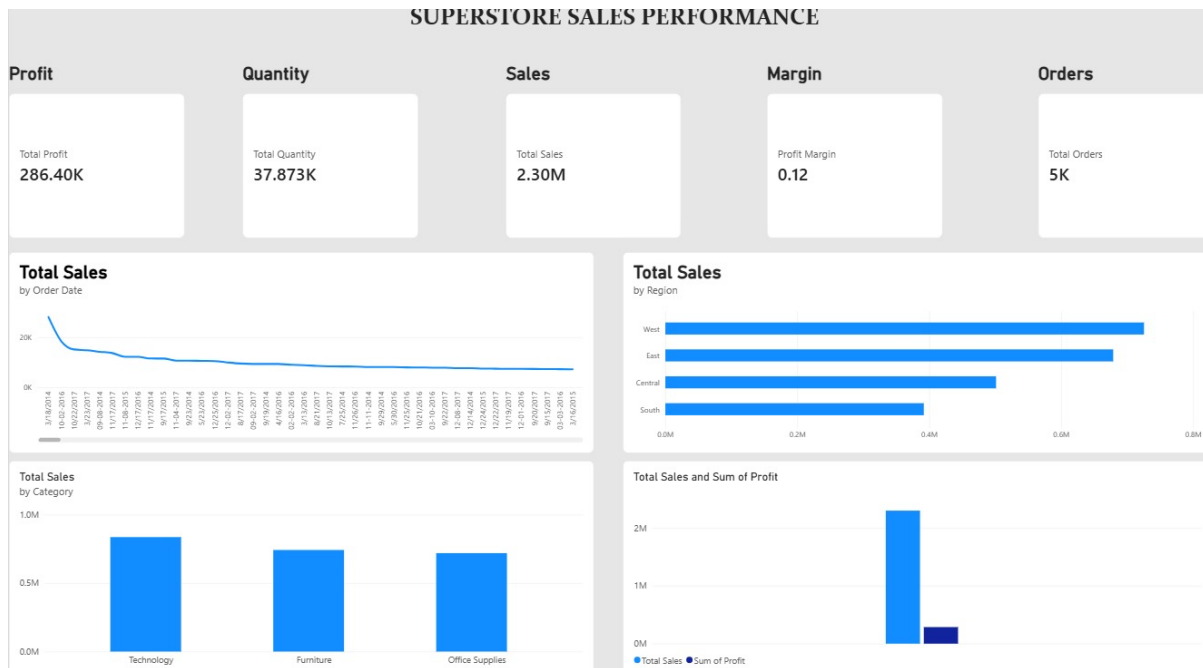
```
1 Total Sales = SUM('Sample_Superstore'[Sales])
1 Total Profit = SUM('Sample_Superstore'[Profit])
1 Total Quantity = SUM('Sample_Superstore'[Quantity])
1 Total Orders = DISTINCTCOUNT('Sample_Superstore'[Order ID])
1 Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
```

Fig 3: DAX measures created for Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin

Step 5: Dashboard Development – Overview Page

- Built the first dashboard page, titled "Superstore Sales Performance", containing:
- KPI cards for Profit, Quantity, Sales, Margin, and Orders.
- A line chart showing Total Sales by Order Date.

- A bar chart showing Total Sales by Region.
- A column chart showing Total Sales by Category.
- A combo chart showing Total Sales and Sum of Profit together.



Step 6: Dashboard Development – Detailed Analysis Page

- Built a second dashboard page containing:
- A bar chart showing Total Sales by Sub-Category.
- A bar chart showing Sum of Profit by Region.
- A bar chart showing Total Profit by Product Name (top contributors).
- A bar chart showing Total Sales by Product Name (top contributors).

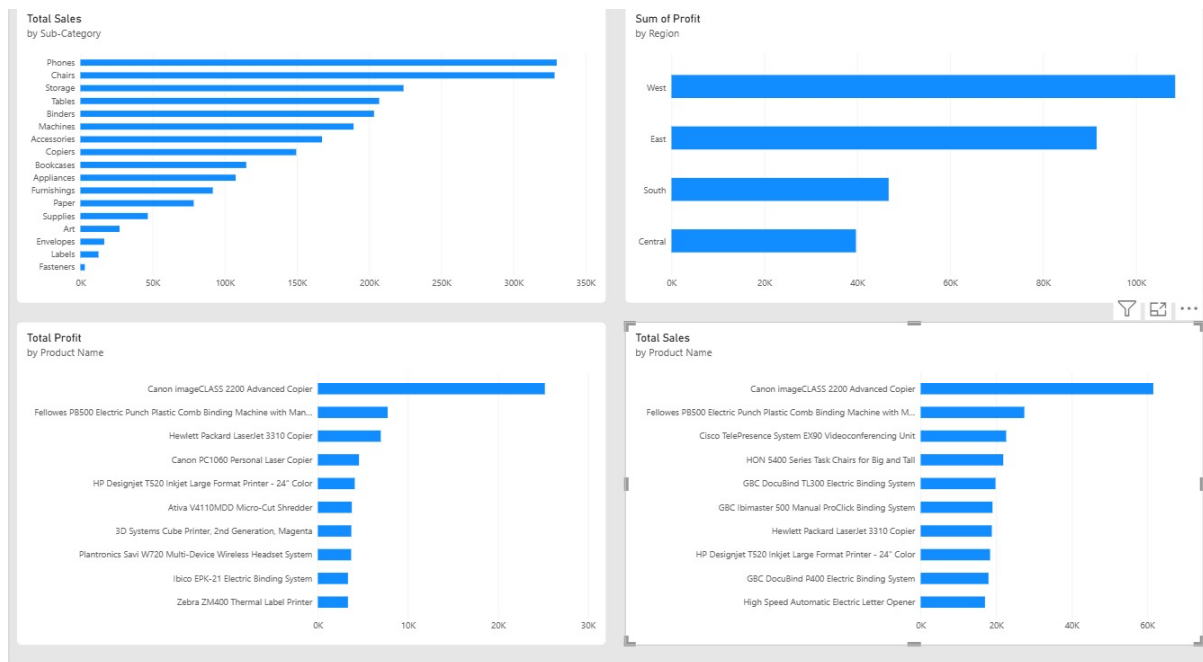


Fig 5: Dashboard – Detailed Analysis page (Sub-Category, Region profit, and Product-wise breakdowns)

Step 7: Dashboard Formatting

- Added the title "Superstore Sales Performance" at the top of the overview page.
- Arranged KPI cards and charts in a structured, easy-to-scan layout across both pages.
- Formatted Sales, Profit, and Quantity values with appropriate number formatting (M for millions, K for thousands).
- Adjusted chart titles and visual sizing for clarity and consistency across both dashboard pages.

Step 8: Analysis and Insights

- Overall business performance: Total Sales of 2.30M, Total Profit of 286.40K, Total Quantity of 37.873K units sold, an overall Profit Margin of 0.12, and 5K total orders.
- Regional performance: the West region generated the highest sales, followed by East, South, and Central.
- Category performance: Technology recorded the highest total sales, closely followed by Furniture and Office Supplies.
- Sub-category performance: Phones and Chairs were the top-selling sub-categories, while Fasteners and Labels contributed the least.
- Product performance: the Canon imageCLASS 2200 Advanced Copier was the single highest contributor to both total sales and total profit.
- The Total Sales by Order Date chart provided visibility into how sales activity fluctuated across the recorded time period.

Conclusion

This project used the Sample Superstore dataset to analyze retail sales performance across products, categories, and regions using Power BI. DAX measures were created to calculate Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin, forming the foundation for a two-page interactive dashboard. The dashboard presents overall business performance through KPI cards, regional and category-wise breakdowns, and product-level analysis, enabling management to quickly identify key business patterns and make data-driven decisions.

Learning Outcomes

Through this project, I learned:

1. Data Collection — how to source and understand a real-world business dataset from Kaggle.
2. Power BI Data Import — how to import and load a flat dataset into Power BI Desktop.
3. Data Exploration — how to verify data structure and quality using Data view.
4. DAX Measures — how to create fundamental DAX measures using SUM, DISTINCTCOUNT, and DIVIDE for business KPIs.
5. Data Visualization — how to build KPI cards, line charts, bar charts, and combo charts.
6. Dashboard Development — how to design a multi-page dashboard covering both a summary overview and a detailed breakdown.
7. Business Analysis — how to interpret sales and profit patterns across regions, categories, and products.
8. Data-Driven Decision Making — practical understanding of how Power BI dashboards support management decision-making.